Midterm Report

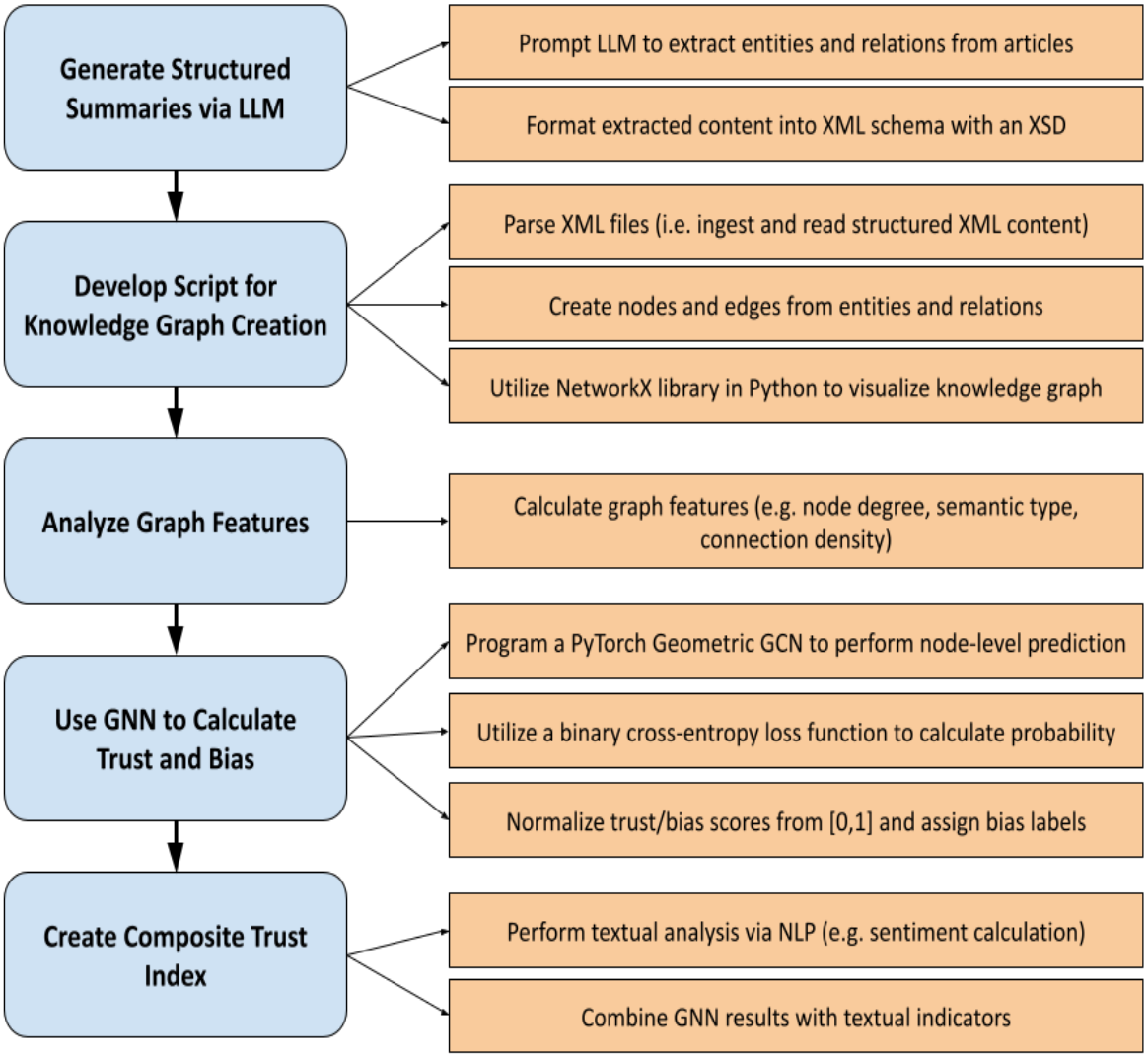
Project 3 — Bias and Trust Analysis in Healthcare AI Summaries

Topic: Alzheimer's Disease

Articles →

- "2024 Alzheimer's Disease Facts and Figures." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*, U.S. National Library of Medicine, 2024, https://pubmed.ncbi.nlm.nih.gov/38689398/.
- Altuna, Miren, et al. "Stepwise Approach to Alzheimer's Disease Diagnosis in Primary Care Using Cognitive Screening, Risk Factors, Neuroimaging and Plasma Biomarkers." *Nature News*, Nature Publishing Group, 27 Aug. 2025, https://ww.nature.com/articles/s41598-025-17394-3.
- Parums, Dinah V. "A Review of the Current Status of Disease-Modifying Therapies and Prevention of Alzheimer's Disease." *Medical Science Monitor : International Medical Journal of Experimental and Clinical Research*, U.S. National Library of Medicine, 13 May 2024, https://pmc.ncbi.nlm.nih.gov/articles/PMC11097689/
- Zhang, Jifa, et al. "Recent Advances in Alzheimer's Disease: Mechanisms, Clinical Trials and New Drug Development Strategies." *Nature News*, Nature Publishing Group, 23 Aug. 2024, https://www.nature.com/articles/s41392-024-01911-3#Abs1.

Project Pipeline Diagram →



LLM-Generated Structured Summaries →

Article 1 (“2024 Alzheimer’s Facts and Figures”):

```
<?xml version="1.0" encoding="UTF-8"?>
<ArticleKnowledge>
  <Metadata>
    <Title>2024 Alzheimer’s Disease Facts and Figures</Title>
    <Journal>Alzheimer’s & Dementia</Journal>
    <Year>2024</Year>
    <PMID>38689398</PMID>
    <Source>PubMed</Source>
    <Focus>Public health impact, prevalence, caregiving, workforce, costs</Focus>
  </Metadata>

  <Entities>
    <Disease id="D1" name="Alzheimer's disease" abbreviation="AD" />
    <Condition id="C1" name="Alzheimer’s dementia" />
    <Condition id="C2" name="Dementia" />
    <Condition id="C3" name="Cognitive impairment" />
    <Disease id="D2" name="COVID-19" />

    <HealthMetric id="H1" type="prevalence" unit="individuals" value="6.9 million" year="2024">
      Americans age 65 and older living with Alzheimer’s dementia
    </HealthMetric>
    <HealthMetric id="H2" type="projection" unit="individuals" value="13.8 million" year="2060">
      Projected number of Americans age 65+ with Alzheimer’s dementia
    </HealthMetric>
    <HealthMetric id="H3" type="mortality" unit="deaths" value="119,399" year="2021">
      Deaths officially recorded from Alzheimer’s disease
    </HealthMetric>
    <HealthMetric id="H4" type="mortality_rank" rank="7">
      Alzheimer’s was the seventh-leading cause of death in the U.S. in 2020–2021
    </HealthMetric>
    <HealthMetric id="H5" type="mortality_rank" rank="5">
      Alzheimer’s remains the fifth-leading cause of death among Americans age 65 and older
    </HealthMetric>
    <HealthMetric id="H6" type="mortality_trend" change="increase_140_percent" years="2000-2021">
      Reported deaths from AD increased more than 140%
    </HealthMetric>

    <CaregiverEntity id="CG1" type="unpaid" population="11 million">
      Family members and other unpaid caregivers (2023)
    </CaregiverEntity>
    <CaregiverMetric id="CGM1" type="care_hours" value="18.4 billion" year="2023">
      Hours of care provided to people with Alzheimer’s or other dementias
    </CaregiverMetric>
    <EconomicMetric id="E1" type="value_unpaid_care" value="$346.6 billion" year="2023">
      Estimated economic value of unpaid caregiving
    </EconomicMetric>
    <WorkforceEntity id="W1" type="health_care">
      Paid health care and community-based dementia workforce
    </WorkforceEntity>
    <WorkforceMetric id="W2" type="shortage" trend="increasing">
      Growing shortages across dementia care workforce
    </WorkforceMetric>

    <EconomicMetric id="E2" type="medicare_cost_ratio" value="3x">
      Medicare payments for beneficiaries with AD vs. without
    </EconomicMetric>
    <EconomicMetric id="E3" type="medicaid_cost_ratio" value="22x">
      Medicaid payments for beneficiaries with AD vs. without
    </EconomicMetric>
    <EconomicMetric id="E4" type="total_cost" value="$360 billion" year="2024">
      Estimated total payments for health care, long-term care, and hospice
```

```
    </EconomicMetric>

    <ProgramEntity id="P1" type="care_model">
      Targeted programs and care delivery models for dementia workforce training
    </ProgramEntity>
    <ProfessionalEntity id="NP1" type="non-physician_healthcare_professional">
      Non-physician professionals facilitating dementia care navigation
    </ProfessionalEntity>
    <SurveyEntity id="S1" type="caregiver_and_healthcare_worker_survey">
      Surveys exploring experiences, challenges, and perceptions of dementia care navigation
    </SurveyEntity>
  </Entities>

  <Relations>
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    <Relation subject="C1" predicate="is_type_of" object="C2" />
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    <Relation subject="D1" predicate="projected_to_reach" object="H2" />
    <Relation subject="D1" predicate="has_mortality" object="H3" />
    <Relation subject="D1" predicate="ranked_as" object="H4" />
    <Relation subject="D1" predicate="ranked_as" object="H5" />
    <Relation subject="D1" predicate="mortality_trend" object="H6" />
    <Relation subject="D2" predicate="co_occurred_with" object="D1" />

    <Relation subject="CG1" predicate="provides_care_for" object="D1" />
    <Relation subject="CG1" predicate="provides_care_hours" object="CGM1" />
    <Relation subject="CG1" predicate="contributes_value" object="E1" />
    <Relation subject="W1" predicate="cares_for" object="D1" />
    <Relation subject="W1" predicate="faces" object="W2" />
    <Relation subject="P1" predicate="addresses" object="W2" />

    <Relation subject="D1" predicate="increases_costs_for" object="E2" />
    <Relation subject="D1" predicate="increases_costs_for" object="E3" />
    <Relation subject="D1" predicate="total_cost_estimate" object="E4" />

    <Relation subject="NP1" predicate="supports" object="care_navigation" />
    <Relation subject="S1" predicate="examines" object="NP1" />
    <Relation subject="S1" predicate="examines" object="CG1" />
  </Relations>
</ArticleKnowledge>
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Article 2 (“Stepwise Approach to Alzheimer’s Disease Diagnosis …”):

```xml
<?xml version="1.0" encoding="UTF-8"?>
<ArticleKnowledge>
  <Metadata>
    <Title>Stepwise approach to Alzheimer’s disease diagnosis in primary care using cognitive screening, risk factors, neuroimaging and plasma biomarkers</Title>
    <Journal>Scientific Reports</Journal>
    <Year>2025</Year>
    <DOI>10.1038/s41598-025-17394-3</DOI>
    <Source>Nature</Source>
    <Focus>Early detection of Alzheimer’s disease via a scalable multimodal framework in primary care</Focus>
  </Metadata>

  <Entities>
    <Disease id="D1" name="Alzheimer’s disease" abbreviation="AD" />
    <Condition id="C1" name="Mild cognitive impairment" abbreviation="MCI" />
    <HealthMetric id="H1" type="prevalence" unit="individuals" value="55.2 million" year="2022" region="Global" />
    <HealthMetric id="H2" type="prevalence" unit="percent" value="2.9" year="2022" region="Southeast Asia" />
    <HealthMetric id="H3" type="prevalence" unit="percent" value="6.5" year="2022" region="Europe" />
    <HealthMetric id="H4" type="projection" unit="individuals" value="78 million" year="2030" region="Global" />
    <Test id="T1" name="MMSE" full_name="Mini-Mental State Examination" />
    <Test id="T2" name="M@T" full_name="Memory Alteration Test" />
    <Test id="T3" name="Fototest" />
    <Test id="T4" name="AD8" />
    <Biomarker id="B1" name="p-tau181" />
    <Biomarker id="B2" name="Aβ42/40 ratio" />
    <Biomarker id="B3" name="GFAP" />
    <Biomarker id="B4" name="NfL" />
    <ImagingMethod id="I1" name="MRI" />
    <GeneticFactor id="G1" name="APOE ε4 allele" />
    <RiskFactor id="R1" name="Age ≥ 60 years" />
    <RiskFactor id="R2" name="Cognitive screening positive" />
  </Entities>

  <Relations>
    <Relation subject="D1" predicate="detected_by" object="T1" />
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    <Relation subject="D1" predicate="detected_by" object="T4" />
    <Relation subject="D1" predicate="associated_with_biomarker" object="B1" />
    <Relation subject="D1" predicate="associated_with_biomarker" object="B2" />
    <Relation subject="D1" predicate="associated_with_biomarker" object="B3" />
    <Relation subject="D1" predicate="associated_with_biomarker" object="B4" />
    <Relation subject="D1" predicate="diagnosed_with_imaging" object="I1" />
    <Relation subject="D1" predicate="risk_factor" object="G1" />
    <Relation subject="D1" predicate="risk_factor" object="R1" />
    <Relation subject="T3" predicate="performed_in" object="primary_care_setting" />
    <Relation subject="B1" predicate="predicts" object="D1" />
    <Relation subject="B3" predicate="predicts" object="D1" />
    <Relation subject="G1" predicate="modulates_risk_of" object="D1" />
  </Relations>
</ArticleKnowledge>
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Article 3 (“A Review of the Current Status of Disease-Modifying Therapies …”):

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<?xml version="1.0" encoding="UTF-8"?>
<ArticleKnowledge>
  <Metadata>
    <Title>A Review of the Current Status of Disease-Modifying Therapies and Prevention of Alzheimer’s Disease</Title>
    <Journal>Medical Science Monitor</Journal>
    <Year>2024</Year>
    <PMCID>PMC11097689</PMCID>
    <PMID>38736218</PMID>
    <Source>PMC (PubMed Central)</Source>
    <Focus>Disease-modifying therapies and prevention of Alzheimer’s disease</Focus>
  </Metadata>

  <Entities>
    <Disease id="D1" name="Alzheimer’s disease" abbreviation="AD" />
    <Condition id="C1" name="Dementia" />
    <Condition id="C2" name="Mild cognitive impairment" abbreviation="MCI" />
    <Disease id="D2" name="Lewy body dementia" />
    <Disease id="D3" name="Vascular dementia" />
    <Disease id="D4" name="Frontotemporal dementia" />
    <Biomarker id="B1" name="amyloid β" />
    <Biomarker id="B2" name="tau protein" />
    <Biomarker id="B3" name="amyloid β oligomers" />
    <ImagingMethod id="I1" name="PET amyloid imaging" />
    <ImagingMethod id="I2" name="tau PET imaging" />
    <ImagingMethod id="I3" name="MRI" />
    <ImagingMethod id="I4" name="FDG-PET" />
    <Test id="T1" name="Mini-Mental State Examination" abbreviation="MMSE" />
    <Test id="T2" name="Montreal Cognitive Assessment" abbreviation="MoCA" />
    <Test id="T3" name="Clinical Dementia Rating" abbreviation="CDR" />
    <Drug id="DR1" name="Aducanumab" />
    <Drug id="DR2" name="Cholinesterase inhibitors" />
    <Drug id="DR3" name="Memantine" />
    <Drug id="DR4" name="Selegiline" />
    <Drug id="DR5" name="Vitamin E (alpha-tocopherol)" />
    <ClinicalTrialMetric id="CT1" type="pipeline_count" value="187">Number of clinical trials in progress for Alzheimer’s disease therapies</ClinicalTrialMetric>
    <RiskFactor id="R1" name="Diabetes mellitus" />
    <RiskFactor id="R2" name="Air pollution (nitrogen dioxide exposure)" />
    <RiskFactor id="R3" name="Frequency of alcohol intake" />
    <RiskFactor id="R4" name="Low education" />
    <RiskFactor id="R5" name="Hearing impairment" />
    <RiskFactor id="R6" name="Obesity" />
    <RiskFactor id="R7" name="Smoking" />
    <RiskFactor id="R8" name="Depression" />
    <RiskFactor id="R9" name="Physical inactivity" />
    <RiskFactor id="R10" name="Social isolation" />
    <RiskFactor id="R11" name="Traumatic brain injury" />
    <Gene id="G1" name="APOE" allele="ε4" />
  </Entities>

  <Relations>
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    <Relation subject="D1" predicate="diagnosed_by" object="I1" />
    <Relation subject="D1" predicate="diagnosed_by" object="I2" />
    <Relation subject="D1" predicate="diagnosed_by" object="T1" />
    <Relation subject="D1" predicate="diagnosed_by" object="T2" />
    <Relation subject="D1" predicate="diagnosed_by" object="T3" />
    <Relation subject="DR1" predicate="targets" object="B1" />
    <Relation subject="DR2" predicate="symptom_relief_for" object="D1" />
    <Relation subject="DR3" predicate="symptom_relief_for" object="D1" />
    <Relation subject="DR4" predicate="evaluated_for" object="D1" />
    <Relation subject="DR5" predicate="evaluated_for" object="D1" />
    <Relation subject="D1" predicate="undergoing_clinical_trials" object="CT1" />
    <Relation subject="R1" predicate="increases_risk_of" object="D1" />
    <Relation subject="R2" predicate="increases_risk_of" object="D1" />
    <Relation subject="R3" predicate="increases_risk_of" object="D1" />
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    <Relation subject="G1" predicate="genetic_risk_for" object="D1" />
    <Relation subject="D1" predicate="preventable_by_modifying" object="R1" />
    <Relation subject="D1" predicate="preventable_by_modifying" object="R2" />
    <Relation subject="D1" predicate="preventable_by_modifying" object="R3" />
    <Relation subject="DR1" predicate="has_limitation" object="cost_and_availability" />
    <Relation subject="DR1" predicate="has_limitation" object="ARIA_side_effects" />
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</ArticleKnowledge>
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Article 4 (“Recent Advances in Alzheimer’s Disease: Mechanisms …”)

```
<?xml version="1.0" encoding="UTF-8"?>
<ArticleKnowledge>
  <Metadata>
    <Title>Emerging Insights into Alzheimer’s Disease Pathogenesis and Drug Development</Title>
    <Journal>Signal Transduction and Targeted Therapy</Journal>
    <Year>2024</Year>
    <DOI>10.1038/s41392-024-01911-3</DOI>
    <Source>Nature</Source>
    <Focus>Pathogenesis, biomarkers, clinical trials, and emerging drug development strategies for Alzheimer’s disease</Focus>
  </Metadata>

  <Entities>
    <Disease id="D1" name="Alzheimer's disease" abbreviation="AD" />
    <Condition id="C1" name="Dementia" />
    <Condition id="C2" name="Mild cognitive impairment" abbreviation="MCI" />
    <HealthMetric id="H1" type="prevalence" unit="individuals" value="55.2 million" year="2022" region="Global" />
    <HealthMetric id="H2" type="prevalence" unit="percent" value="2.9" year="2022" region="Southeast Asia" />
    <HealthMetric id="H3" type="prevalence" unit="percent" value="6.5" year="2022" region="Europe" />
    <HealthMetric id="H4" type="projection" unit="individuals" value="78 million" year="2030" region="Global" />
    <HealthMetric id="H5" type="incidence" unit="per 100,000" value="1275" year="2022" region="US age 65+" />
    <Pathway id="P1" name="Cholinergic hypothesis" />
    <Pathway id="P2" name="Amyloid hypothesis" />
    <Pathway id="P3" name="Tau protein hypothesis" />
    <Pathway id="P4" name="Inflammatory hypothesis" />
    <Pathway id="P5" name="Oxidative stress hypothesis" />
    <Pathway id="P6" name="Metal ion hypothesis" />
    <Pathway id="P7" name="Glutamate excitotoxicity hypothesis" />
    <Pathway id="P8" name="Microbiota–gut–brain axis hypothesis" />
    <Pathway id="P9" name="Autophagy dysfunction hypothesis" />
    <Biomarker id="B1" name="Amyloid-β (Aβ) plaques" />
    <Biomarker id="B2" name="Neurofibrillary tangles (NFTs)" />
    <Drug id="DR1" name="Aducanumab" approval="FDA" />
    <Drug id="DR2" name="Lecanemab" approval="FDA" />
    <Drug id="DR3" name="Donanemab" status="Under review" />
    <Drug id="DR4" name="Tacrine" status="Withdrawn" />
    <Drug id="DR5" name="Donepezil" />
    <Drug id="DR6" name="Rivastigmine" />
    <Drug id="DR7" name="Galantamine" />
    <Drug id="DR8" name="Memantine" />
    <Drug id="DR9" name="Namzaric" />
    <Drug id="DR10" name="Sodium oligomannate (GV-971)" />
    <DrugApproach id="DA1" name="Selective inhibitors" />
    <DrugApproach id="DA2" name="Dual-target inhibitors" />
    <DrugApproach id="DA3" name="Allosteric modulators" />
    <DrugApproach id="DA4" name="Covalent inhibitors" />
    <DrugApproach id="DA5" name="Proteolysis-targeting chimeras (PROTACs)" />
    <DrugApproach id="DA6" name="Protein–protein interaction (PPI) modulators" />
    <RiskFactor id="R1" name="Aging" />
    <RiskFactor id="R2" name="Genetic predisposition" />
    <RiskFactor id="R3" name="Environmental factors" />
    <RiskFactor id="R4" name="Hypercholesterolemia" />
    <RiskFactor id="R5" name="Hypertension" />
    <RiskFactor id="R6" name="Diabetes" />
    <RiskFactor id="R7" name="Obesity" />
    <RiskFactor id="R8" name="Depression" />
    <RiskFactor id="R9" name="Cardiovascular disease" />
  </Entities>
```

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  <Relations>
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    <Relation subject="D1" predicate="characterised_by" object="B2" />
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    <Relation subject="DR2" predicate="potentially_modifies" object="D1" />
    <Relation subject="DR3" predicate="potentially_modifies" object="D1" />
    <Relation subject="DR4" predicate="withdrawn_due_to" object="hepatotoxicity" />
    <Relation subject="DR5" predicate="symptomatic_relief_for" object="D1" />
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    <Relation subject="DR9" predicate="symptomatic_relief_for" object="D1" />
    <Relation subject="DR10" predicate="disease_modifying_candidate_for" object="D1" />
    <Relation subject="DA1" predicate="applied_to" object="D1" />
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    <Relation subject="DA3" predicate="applied_to" object="D1" />
    <Relation subject="DA4" predicate="applied_to" object="D1" />
    <Relation subject="DA5" predicate="applied_to" object="D1" />
    <Relation subject="DA6" predicate="applied_to" object="D1" />
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    <Relation subject="R3" predicate="increases_risk_of" object="D1" />
    <Relation subject="R4" predicate="modulates_progression_of" object="D1" />
    <Relation subject="R5" predicate="modulates_progression_of" object="D1" />
    <Relation subject="R6" predicate="modulates_progression_of" object="D1" />
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    <Relation subject="R9" predicate="modulates_progression_of" object="D1" />
  </Relations>
</ArticleKnowledge>
```

Preliminary Knowledge Graph Creation Script →

Knowledge_Graph.py > ...

```python
import xml.etree.ElementTree as ET
import networkx as nx
import matplotlib.pyplot as plt

Tabnine | Edit | Test | Explain | Document
def parse_xml_to_graph(xml_file):
    tree = ET.parse(xml_file)
    root = tree.getroot()

    G = nx.DiGraph()

    # Add entities
    for entity in root.findall('./Entities/*'):
        entity_id = entity.attrib.get('id')
        entity_type = entity.tag
        entity_name = entity.attrib.get('name')
        G.add_node(entity_id, type=entity_type, name=entity_name)

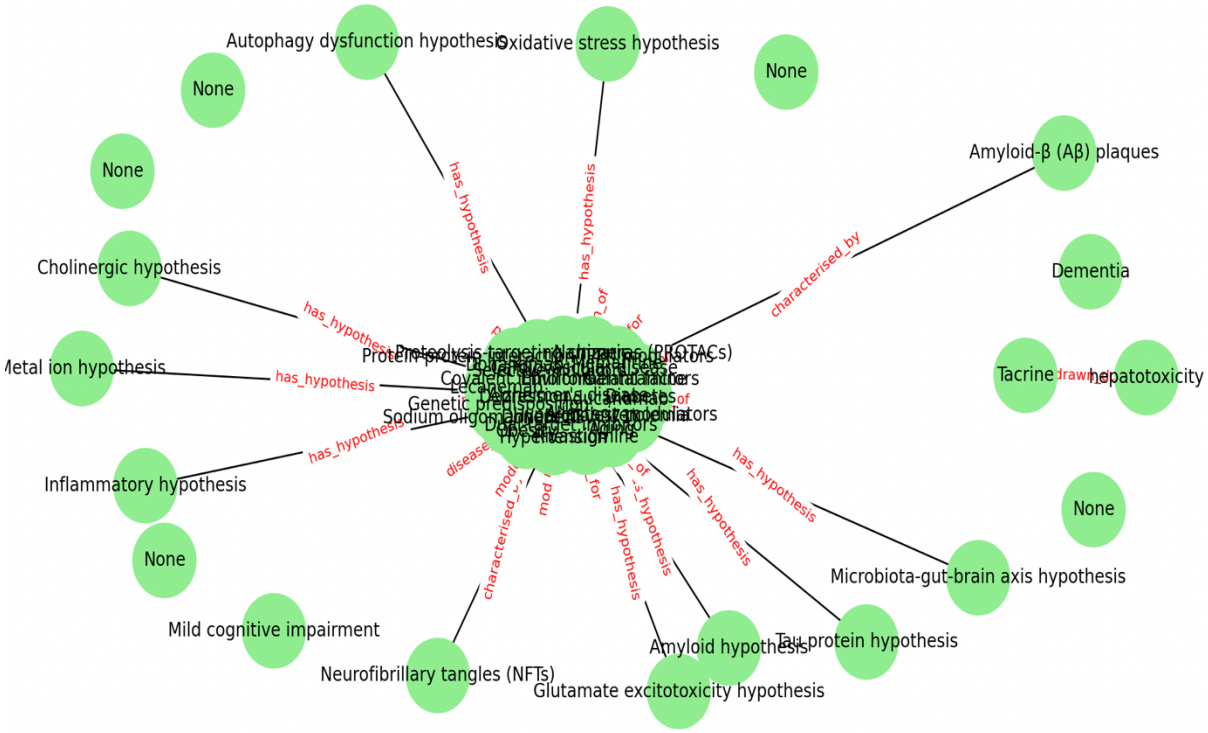
    # Add relations
    for relation in root.findall('./Relations/Relation'):
        subj = relation.attrib.get('subject')
        obj = relation.attrib.get('object')
        pred = relation.attrib.get('predicate')
        G.add_edge(subj, obj, predicate=pred)

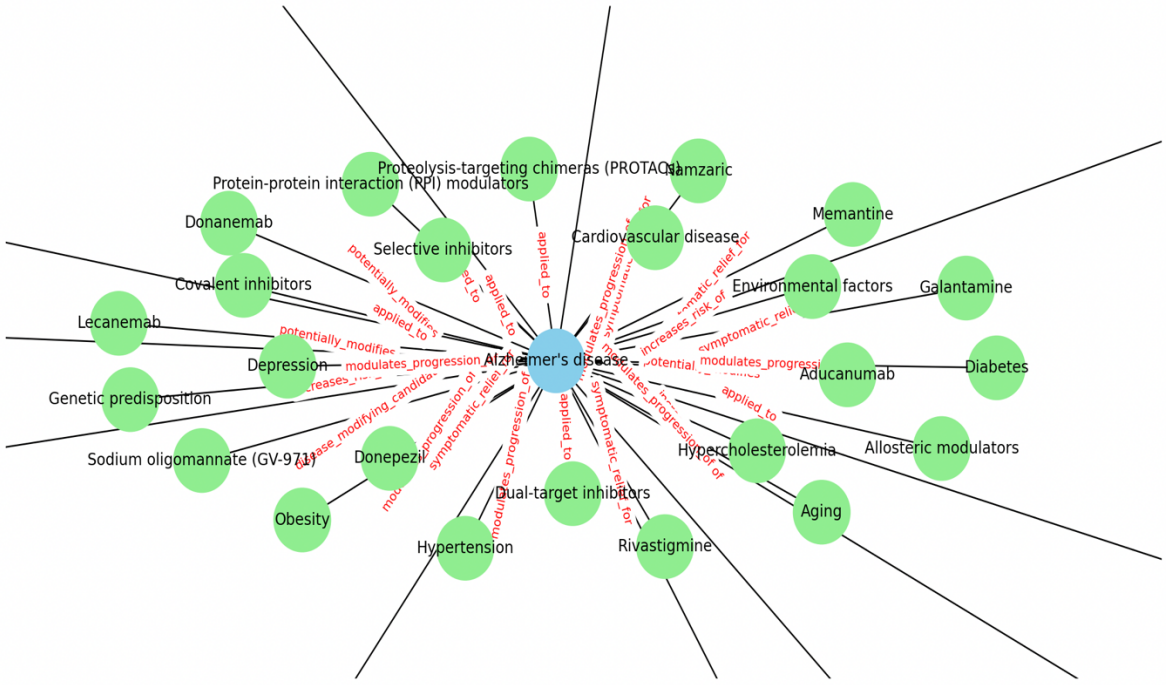
    return G

Tabnine | Edit | Test | Explain | Document
def visualize_graph(G):
    pos = nx.spring_layout(G, seed=42)
    plt.figure(figsize=(12, 12))
    node_labels = {n: G.nodes[n].get('name', str(n)) for n in G.nodes()}
    node_colors = [
    'skyblue' if G.nodes[n].get('type', '') == 'Disease' else 'lightgreen'
    for n in G.nodes()
    ]
    nx.draw_networkx_nodes(G, pos, node_color=node_colors, node_size=1500)
    nx.draw_networkx_labels(G, pos, labels=node_labels, font_size=10)
    edge_labels = nx.get_edge_attributes(G, 'predicate')
    nx.draw_networkx_edges(G, pos, arrowstyle='->', arrowsize=20)
    nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels, font_color='red', font_size=8)
    plt.axis('off')
    plt.show()

if __name__ == "__main__":
    xml_file = "AD_Pathogenesis_Drug_Development_2024.xml"  # test for each XML file
    G = parse_xml_to_graph(xml_file)
    visualize_graph(G)
```

Sample Knowledge Graph →





Next Steps →

- Develop XML Schema Definition to improve Knowledge Graph
- Develop a script to create a knowledge graph merging content from all four articles
- Program GNN Module to calculate trust/bias scores
- Calculate textual indicator metrics (e.g. sentiment analysis)
- Create composite trust index merging GNN results with NLP metrics